

MINGYUN GUAN

Seattle, WA · guangoku@gmail.com · mingsplayground.com · linkedin.com/in/mingyun-guan-founder-data

Data Platform & ML Engineer · Founder, Atolla Ocean · Ex-Amazon Tech Lead

Data and ML engineer with 10+ years building large-scale data platforms and the ML systems that run on them, from pipelines to models in production. At Amazon, IC tech lead across three scopes: a full-stack ML product as its founding engineer, a centralized data platform team, and the ML data platform on restricted-tier customer data. Also led privacy and security reviews across orgs. Strongest at zero-to-one work out of ambiguity, then scaling it, translating between engineers, scientists and product. Now founder of Atolla Ocean, a multimodal video AI product with its own MLOps and evals.

EXPERIENCE

Founder · Atolla Ocean

Jan 2026 – Present · Seattle, WA

Turns divers' video and photos into highlight clips and a digital aquarium. Built from zero and used by other divers. Founder side: product direction, positioning and user interviews. Founder Institute graduate.

- Own a multi-stage batch pipeline over multimodal dive video: frame filtering, species ID by embedding retrieval on research models (BioCLIP, Grounding DINO), LLM enrichment, video enhancement (color correction, stabilization), highlight clips.
- Built the eval and data loop: a labeled held-out eval set and harness comparing models on performance, latency and cost, plus confidence triage and a review tool where corrections become labels for per-stage evals.
- Own every other layer: model routing, backend, front end, deployment and inference cost.

Sole Engineer · CharityBox (益盒)

Jan 2025 – Present · Remote, part-time

Effective-giving platform in China, incubated by MiraclePlus (China's YC); 1,500+ members pledge 1% of their income to vetted charities. Sole engineer for the entire tech function, full-stack, and driving its AI transformation.

- Brought two contractor builds in-house: a new payment system (one-time, group-campaign and special-event donations beside the original subscriptions; Java backend, front end, admin) and an automated operations data sync pipeline that removed about 70% of weekly manual operations effort.
- Built the AI agent workflows and harness that make that pace viable: triage, PRD drafting and delivery run on written procedures, with a shared knowledge base that agents read, code-enforced permission tiers and human gates on production and member-data writes.

Senior Data Engineer (Tech Lead) · Amazon

Jan 2019 – Dec 2024 · Seattle, WA

IC tech lead across three scopes, below. Promoted to DE III and tech lead in Oct 2020.

Full-stack ML product · founding engineer and tech lead of 6 engineers, mostly SDEs

- Took a zero-to-one internal web product with heavy ML from prototype to production, alongside science and PM.
- Architected and built the stack end to end: multi-source ingestion over standard and bulk APIs → knowledge graph with entity resolution → trend detection → search and ranking APIs on OpenSearch serving the web app.
- Owned external data end to end: evaluated vendors with product, then onboarded 5 of them (APIs, Snowflake, S3/Redshift subscriptions) with schemas, metadata and data-quality monitoring.
- Ran its models in production via SageMaker and CI/CD: 5 batch models for knowledge-graph construction and near-real-time ranking models serving the product.

Centralized data platform team · tech lead of 5 data engineers

- Owned data infrastructure for a 200+ person org of strategists, scientists, engineers and economists: operated the warehouse (workload and user management), built new lakehouse systems and self-serve tools, and supported onboarded teams with data-engineering expertise.

Customer Service ML data platform

- Built from zero the ML data platform used to extract data and train models on restricted-tier customer data; worked on the ML-system design as it expanded across the org.
- Built Kinesis streaming and NLP pipelines for contact classification; defect-detection models (AutoGluon, deep learning).
- Built the continuous-training pipeline: Ground Truth labeling, scheduled retraining, golden-set evaluation, drift monitoring.

Across orgs · led privacy and security reviews; hiring loops, mentoring and AWS cost optimization.

Business Intelligence Engineer · Amazon Kids+

Oct 2017 – Dec 2018 · Seattle, WA

- Designed scalable Redshift data models and analytical foundations for product, marketing and finance.
- Built self-serve dashboards for non-technical teams; delivered analyses shaping pricing strategy and UX decisions.

Senior Data Analyst, Machine Learning Platform · Universal McCann Worldwide Feb 2015 – Sep 2017 · New York, NY
Promoted from Data Analyst in 2017 to lead analytics automation efforts.

- Built marketing-mix and Bayesian models to measure media impact and optimize budgets.
- Designed self-serve ETL and data-quality tools: 90% faster workflows, ~80% fewer errors.

SKILLS

AI systems: LLM APIs (Anthropic, OpenAI, Google), multi-model routing and cost accounting, embedding retrieval and reranking, evaluation harnesses, agent tooling (Claude Agent SDK), guardrails and permission design

Governance, privacy & security: Restricted-tier data platform design, privacy and security reviews, data classification and access control, human-gated agent operations

ML data & MLOps: Training and evaluation datasets, labeling and annotation workflows (SageMaker Ground Truth), multimodal video and image pipelines, MLOps (SageMaker, CI/CD), model performance monitoring

Data platform: Large-scale batch and streaming pipelines (Kinesis), lakehouse architecture, workflow orchestration (Step Functions, SageMaker Pipelines), multi-source ingestion, schema and metadata design, data-quality monitoring, Spark

Cloud & backend: AWS (S3, Glue, Redshift, Athena, Kinesis, Lambda, Step Functions, ECS/Fargate, SageMaker, OpenSearch, CDK), Snowflake, FastAPI, PostgreSQL, SQLAlchemy, async Python, pytest

Languages: Python, SQL, TypeScript, Java, JavaScript

EDUCATION

MA in Statistics · Columbia University	2015 · New York, NY
Bachelor of Economics in Statistics (Minor: Mathematics) · Xiamen University	2013 · Xiamen, China